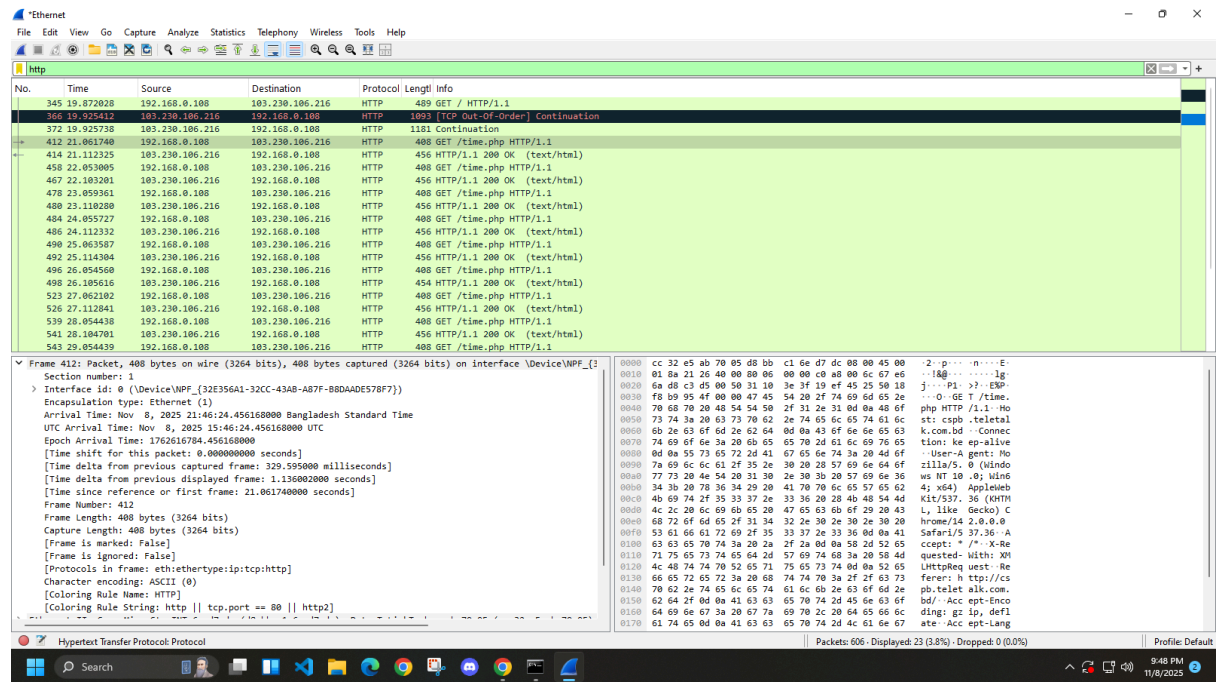
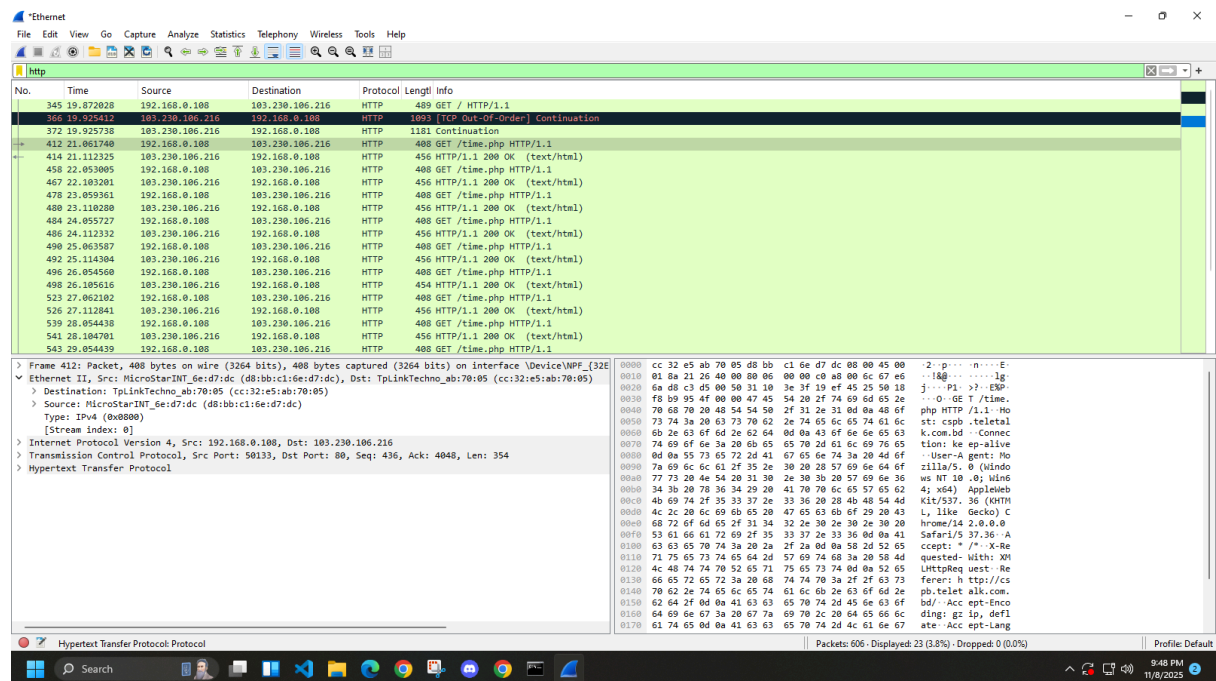


1.



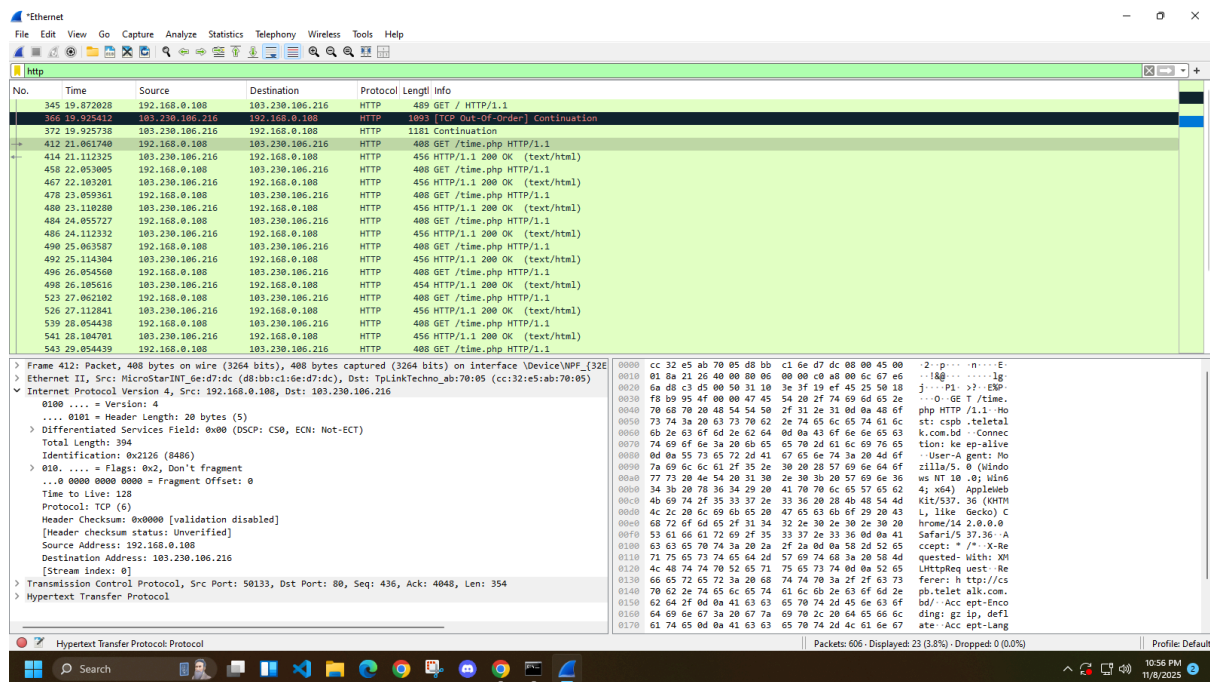
This shows the frame header which is part of the physical layer. We can see the timestamps of the physical layer that show when the frame was detected and that the encapsulation type was Ethernet(1).

2.



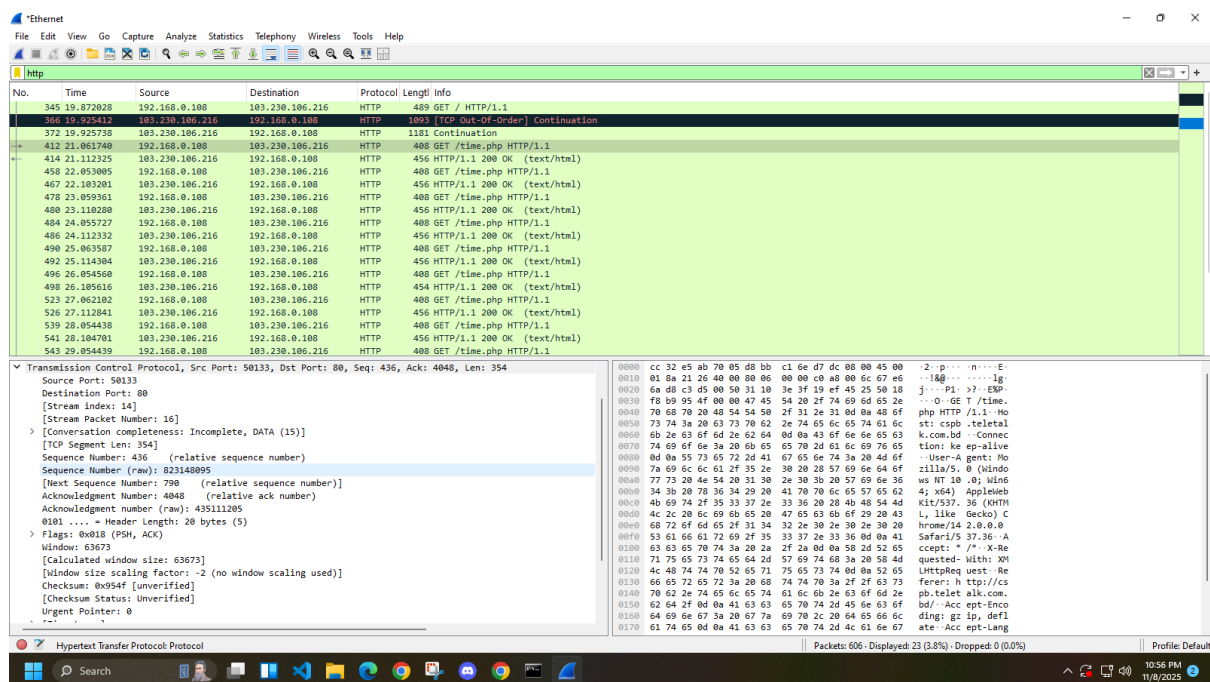
This shows the Ethernet II header which gives the source mac address of the local pc and the destination address which is the router, showing that frame transfer is happening in data link layer.

3.



This shows the Inter Protocol Version 4 header which is in the Network Layer . The header length shows the size of the ip header which is 20 bytes and the total length is 394 bytes which is the total size of the header and payload. Time To Live or TTL shows how many hops the packet can do and the protocol is TCP. it also shows source address and destination address.

4.



This shows the Tranmission Control Protocol header which is part of the Transport Later in the OSI model. There is source port 50133 and destination port 80 which shows system is communicating with web server. The sequence number 436 shows the position of the first

byte in the segment and the acknowledgement number of 4048 shows the confirmation of receiving previously sent bytes. The header length shows the size of the TCP header and the PSH, ACK show that the packet is acknowledging the received data and pushing buffer data both. The window size of 63673 shows how many bytes sender can receive. The checksum 0x954f is used to detect errors in the transmission and urgent pointer 0 means no urgent data is sent.

5.

The image shows a Wireshark packet capture of an HTTP GET request. The packet list pane at the top shows a continuation of a previous packet (No. 345). The packet details pane shows the full HTTP request structure, including the GET method, URI, version, host, connection, user-agent, accept, and accept-encoding headers. The packet bytes pane shows the raw data in hexadecimal and ASCII.

Frame 412: Packet, 400 bytes on wire (3264 bits), 400 bytes captured (3264 bits) on interface Device\NPF... (3264) [Ethernet II, Src: MicroStarIntf6:d7:dc (d8:bb:1c:6e:d7:dc), Dst: TplinkTechno... (cc:32:e5:ab:70:05)]

Internet Protocol Version 4, Src: 192.168.0.108, Dst: 103.230.106.216

Transmission Control Protocol, Src Port: 50133, Dst Port: 80, Seq: 436, Ack: 4048, Len: 354

Hypertext Transfer Protocol

GET /time.php HTTP/1.1

Request Method: GET

Request URI: /time.php

Request Version: HTTP/1.1

Host: cspb.teletalk.com.bd

Connection: keep-alive

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0

Accept: */*

X-Requested-With: XMLHttpRequest

Referer: http://csqb.teletalk.com.bd

Accept-Encoding: gzip, deflate

Accept-Language: en-US,en;q=0.9

[Response in frame: 414]

[Full request URI: http://csqb.teletalk.com.bd/time.php]

This shows the Hypertext Transfer Protocol header which is part of the Application layer in the OSI model. Get/time.php HTTP/1.1 means that an HTTP get request is being sent to get the time.php from server. The request URI shows the page that is being requested and Request version: HTTP/1.1 shows which version of the protocol is being used. The host header shows the domain name of the we server taht is being contacted. The connection: keep-alive makes sure the server keeps TCP connection open for more requests. Encoding: gzip, deflate shows that there is encoding to lessen data size. We can also see that there is Accept-Language:en-US which is preferred language. We can also see referrer which is teletalk site and the URI of the full request.

6.

The screenshot shows Wireshark capturing an HTTP GET request. The packet list on the left shows packet 414 selected. The packet details pane on the right shows the Ethernet II header with source MAC d8:bb:c1:6e:d7:dc and destination MAC e5:ab:70:05:08:00. The packet bytes pane on the right shows the raw data of the frame.

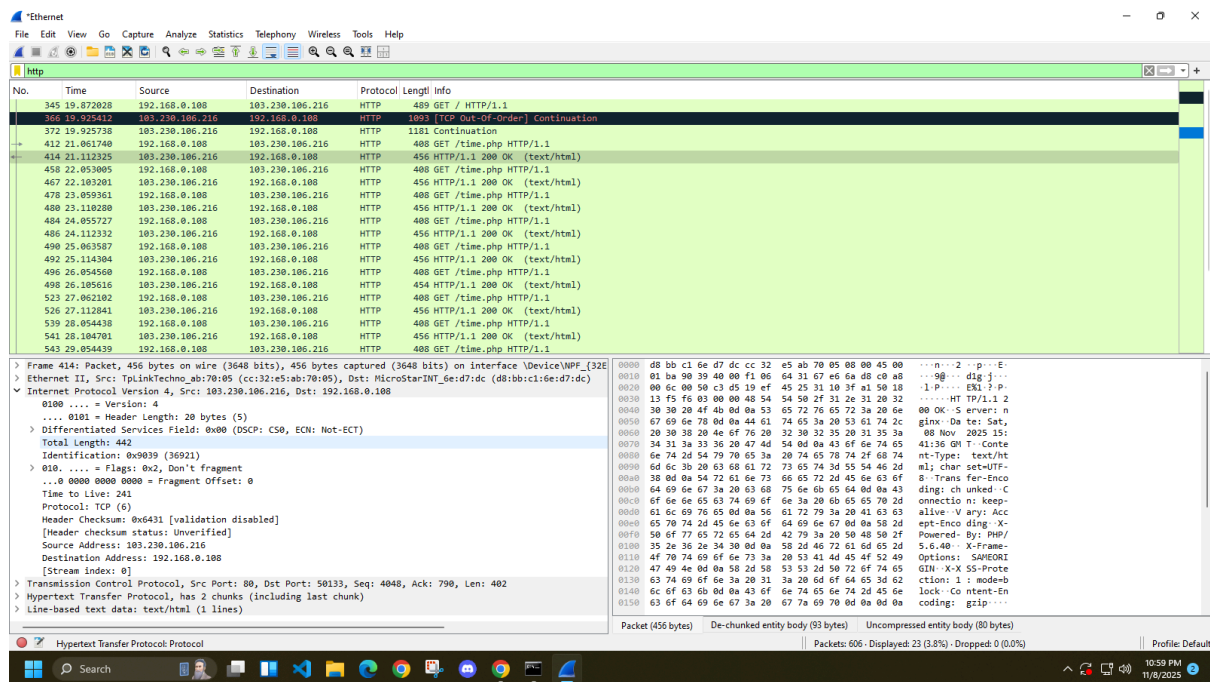
This shows the Frame header from the receive side which is part of Physical Layer. We can see the total frame size which is 456 bytes and the encapsulation type is Ethernet II and also there are timestamps of frame reaching the machine.

7.

The screenshot shows Wireshark capturing an HTTP GET request. The packet list on the left shows packet 414 selected. The packet details pane on the right shows the Ethernet II header with source MAC d8:bb:c1:6e:d7:dc and destination MAC e5:ab:70:05:08:00. The packet bytes pane on the right shows the raw data of the frame.

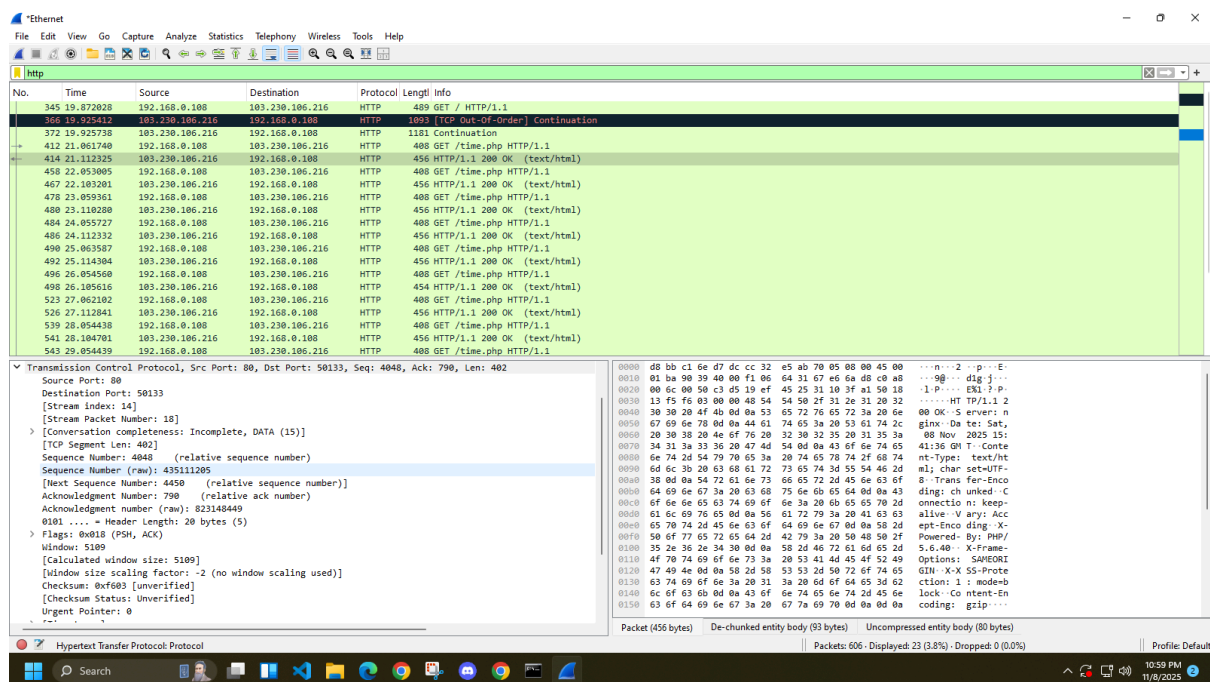
This shows the Ethernet II header which is part of Data Link layer of the OSI model. It shows the source and destination mac addresses which mean that frame is transferred between two networks.

8.



This shows the Internet Protocol Version 4 header which is part of the Network layer in the OSI model. The header length shows the size of the the header and the total length shows the overall size of the packet. TTL shows the lifetime of the packet. The protocol is TCP and the source address and destination address specify the the ip addresses of the sender and the receiver.

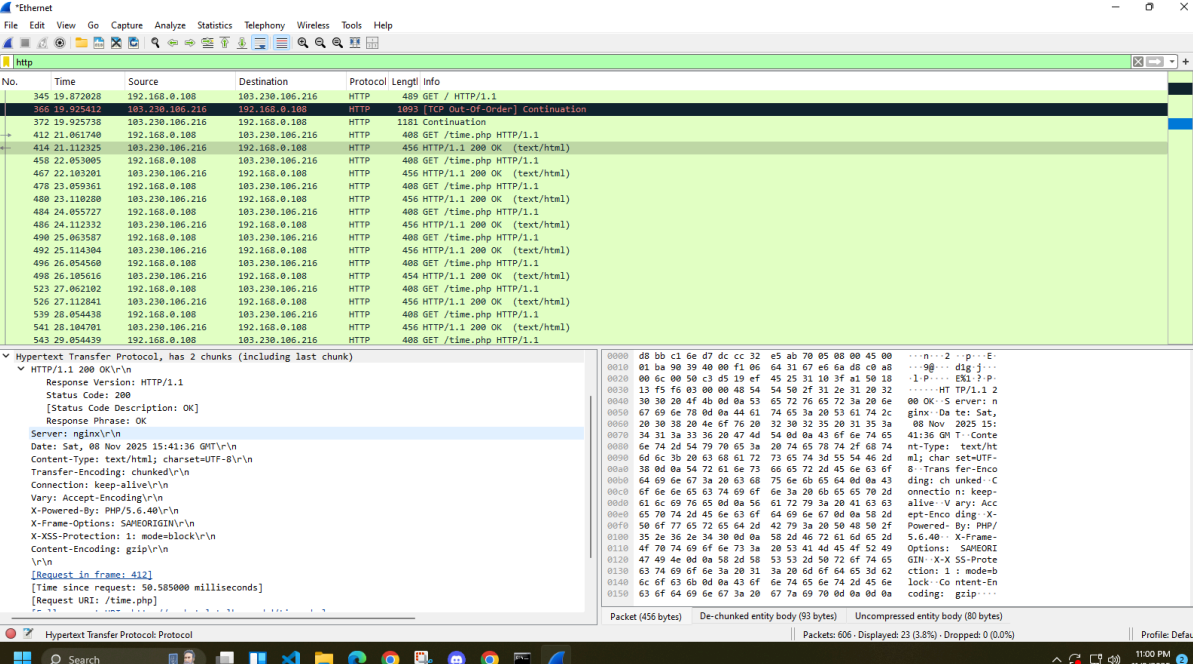
9.



This shows the Transmission Control Protocol header which is part of the Trasport Layer in the OSI model. The Source Port: 80 and Destination Port 50133 show that the

communication starts from the web server and goes to the client. The sequence number 4048 shows the position of the starting byte in the segment. the acknowledgement number 790 confirms that the data is received. The flags PSH and ACK shows that sender does both acknowledgement of the data being received and push the buffer data. Window size 5109 shows the amount of data sender can receive. The checksum detects error in the transmission process. Urgent pointer 0 means there is no urgent data at the moment.

10.



The screenshot shows a Wireshark packet capture of an HTTP 200 OK response. The packet list pane shows a list of packets, with the selected packet being the 4048th byte, which is the start of the response. The packet details pane shows the HTTP header, including the status code 200 OK and the content type text/html; charset=UTF-8. The packet bytes pane shows the raw data of the packet, including the status code and the content type.

This shows the Hypertext Transfer Protocol header which is part of the Application layer of the OSI model. HTTP/1.1 shows the protocol which ensures persistent connections. HTTP/1.1 200 OK means that the request is successful. Here, the content is shown as text/html; charset=UTF-8 which says that the load is an HTML document encoded in UTF-8. Transfer encoding: chunked means that the body is sent in separate segments. Connection: Keep-alive ensures that the session can be reused. Vary: accept encoding means that different versions should be stored based on compression type in the caches. X-Frame-Options and X-Xss-Protection provides security measures against clickjacking and malicious scripts. X-Powered-By: PHP shows that php has been used in backend. Also, content-